Tuning Autovacuum Parameters

Autovacuum runs Vacuum workers per table when threshold values are met. These threshold values are configurable and there are a lot of them for different purposes.

Generally speaking, you may tune Autovacuum so that it runs jobs more frequently, for longer periods of time, and more of them in parallel. The tradeoff is that by running more workers for longer, you'll use more server system resources. These are the same server resources available to your application query workload, so you'll need to carefully make adjustments and monitor the impact.

As your database gets busier with higher rates of UPDATE and DELETE operations, dead row versions accumulate faster. Autovacuum may fall behind and not clean up dead tuples quickly enough before going to sleep again.

When Autovacuum is not keeping up you'll see a symptom of long VACUUM durations where it might run for hours or even days. VACUUM run times are logged in the postgresql.log.

The threshold to trigger an Autovacuum worker for a table is controlled by the parameter autovacuum_vacuum_threshold. This can be adjusted by tuning the autovacuum_scale_factor.

The default value for the scale factor is 0.2. This means that when 20% of the table size is made up of dead tuples VACUUM is triggered.

Many resources recommend reducing autovacuum_scale_factor so that VACUUM is triggered when there are fewer dead tuples. A value of 0.01 (1%) or even less will cause VACUUM to run much more frequently, which may be necessary based on the rate that dead tuples accumulate. As you make these changes, first make the change to specific tables that have a high volume of Updates and Deletes before making the change globally for all tables.

To get more familiar with this, you'll begin to tune Autovacuum for the Rideshare database.

The parameter autovacuum_vacuum_cost_limit[4] specifies a "cost" of work that Autovacuum completes before it stops running. Raising this integer value to specify a higher cost causes Autovacuum to run longer, completing more work.

4. https://postgresqlco.nf/doc/en/param/autovacuum_vacuum_cost_limit/